Short-term synaptic plasticity

Computational Neuroscience
University of Bristol
Adapted from B Mizusaki's slides
M Rule

Learning outcomes:

- Describe the concept, mechanisms of Short-Term Plasticity (STP)
- Be able to **write down** the equations for a simple model of STP
- Be able to describe STP as an information filter.

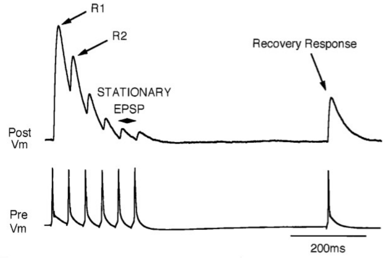
Short-term plasticity

Short-term synaptic plasticity is thought to be involved in fast cognitive processing.

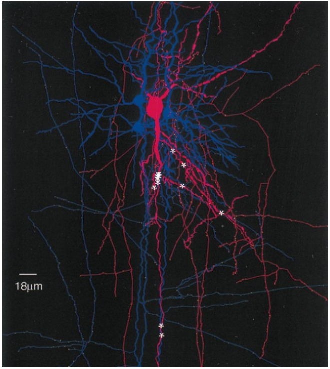
- Similar timescale to electrical dynamics
  - Short-term synaptic facilitation (strength increases with use)
  - Short-term synaptic depression (strength decreases with use)

STP in an experiment

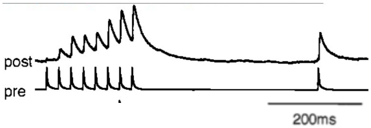
Depression



Tsodyks & Markram, PNAS (1997)



Facilitation



Markram, Wang, & Tsodyks, PNAS (1998)

Short-term depression

Too many vesicles are used before more of them are prepare to be released

- Depletion of the readily releasable pool

Rate of vesicle replenishment limits rate of transmission:

- Low-pass filter

Draw what happens with synaptic vesicles and post-synaptic potentials

Short-term facilitation

AP induces local Ca^{2+} influx in the ***presynaptic*** terminal

- Elevated Ca^{2+} leads to increased release probability
- Ca^{2+} trigger for vesicle fusion nonlinear (release still requires presynaptic spike)

Time of Ca^{2+} availability in the terminal limits period of available facilitation:

- High-pass filter

Draw what happens with synaptic vesicles and post-synaptic potentials

there are other mechanisms but this is what we'll focus on in this course

Short-term depression model

Think of the synaptic efficacy as being determined by the available amount of a resource

$$\begin{aligned}
\dot{R} &= \rho_{\text{rec}}\, I - D \cdot R \cdot \delta(t - t_s) \\
\dot{E} &= -\rho_{\text{inac}}\, E + D \cdot R \cdot \delta(t - t_s) \\
I &= 1 - R - E
\end{aligned}$$

$R \in \{0, 1\}$ fraction "readily" available

- Decrements by dR on each spike
- Recharges from I fraction rate $\rho_{\text{rec}} I$
- $D \in [0, 1] \sim$ portion resource depleted with each spike

$E \in \{0, 1\}$ fraction "exhausted":

- Increments by $D \cdot R$ on each spike
- $E \rightarrow I$ rate $\rho_{\text{inac}} E$

$I \in \{0, 1\}$ fraction "inactive"

- Recycled, but not ready to use
- $I \rightarrow R$ at rate $\rho_{\text{rec}} I$

The lecturer probably needs to explain that $\delta(t - t_s)$ notation

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The Dirac delta notation $q \cdot \delta(t - t_s)$ on the RHS of an ODE is shorthand for a jump condition

- It means that we should add amount q to the integrated variable when we pass through time t_s
- You may also see notation t^- and t^+ to denote that two values may be associated with a single time point before and after the jump.
- It is tempting to assume $\delta(t - t_s)$ can always be viewed as an impulse as in the case of linear ODEs
  - Not always true when the condition generating the impulse comes from the ODE's own state.
  - Writing out the jump conditions explicitly might be more clear.

But, neuroscientists love writing spike (or event) squences as impulse trains $y(t) = \sum_{t_s \in \mathcal{T}} \delta(t - t_s)$.

How should the (R, E, I) modulate synaptic weight?

Here is a first-order exponential decay synapse ($q\sim$ neurotransmitter released)

$$\begin{aligned} g_s(t) &= \bar{g} \cdot s(t) \\ \tau_s \dot{s} &= -s + q \cdot \delta(t - t_s) \end{aligned} \tag{1}$$

We could do this

$$\begin{aligned} g_s(t) &= \bar{g} \cdot s(t) \\ \tau_s \dot{s} &= -s + R \cdot q \cdot \delta(t - t_s) \end{aligned} \tag{2}$$

Can you think of other reasonable choices?

What would we change to get a facilitatory model?

Summary

STP can be modelled as a competition for resources

STP works as a frequency filter

- STF: Low pass
- STD: High pass

Most neurons have synapses that are either mostly facilitating or depressing *end*

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… Quantum Chemistry, Molecular dynamics

Physiological, Quantitative

Biological Realism, Data needed to identify parameters

Molecules	Gillespie, Master Equation
Concentrations	Mass-Action Kinetics
Conductance Models	Hodgkin–Huxley
Spiking Models	Leaky Integrate and Fire
Rate Neurons	Neural Mass/Field Models
Poisson Neurons	Generalized Linear Models
Binary Neurons	McCulloch–Pitts, Hopfield, Perceptron

Computational Efficiency, Mathematical Tractability

Phenomenological, Qualitative

Cognitive Neuroscience, Psychology …